Solving AAS and ASA Triangles with the Law of Sines

Precalculus

Finding the third angle first, then solving AAS and ASA triangles with the law of sines

The tool. $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ (each side is named by the angle *opposite* it; $A+B+C = 180°$). Every figure is drawn to scale from the given data, with the requested quantity marked ?.

- **AAS** — the given side is opposite a given angle: one side–angle pair is already complete, so use the law of sines at once.
- **ASA** — the given side lies *between* the given angles: find the third angle first, then use the law of sines.
- Round side lengths to two decimal places; keep full precision until the last step.

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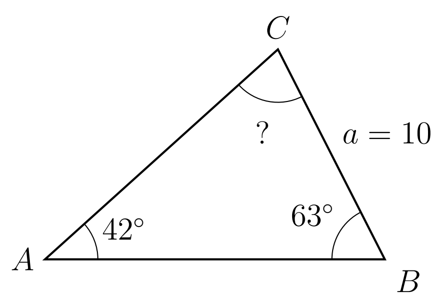
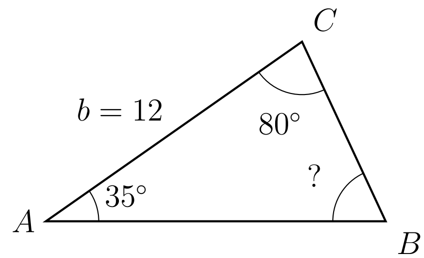
1. Two angles of this triangle are given. Find the third angle C.



Answer: ____________________

2. Find the missing angle B in this triangle. Then state whether the given data is AAS or ASA, and say how you can tell.



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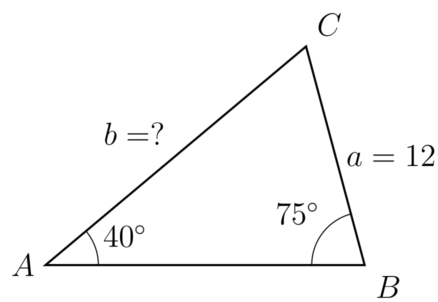
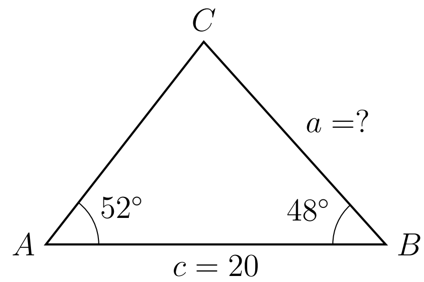
3. **AAS.** Angles A and B are given along with side a, which is opposite A. Use the law of sines to find side b.



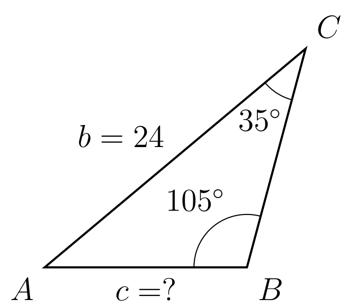
Answer: ________________

4. **ASA.** Here the given side c lies *between* the two given angles, so no side–angle pair is complete yet. Find the third angle first, then find side a.



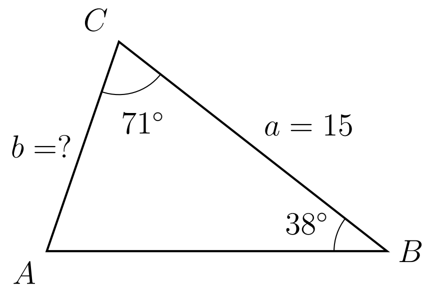
Answer: ____________________

5. **AAS.** Side b is opposite the given angle B. Find side c.



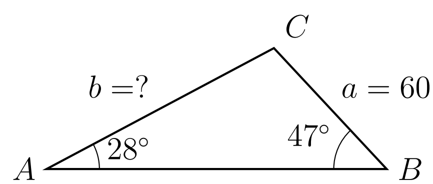
Answer: ____________________

6. ASA. Find side b. Set up the proportion you use before you compute anything, and name which angle you had to find first.



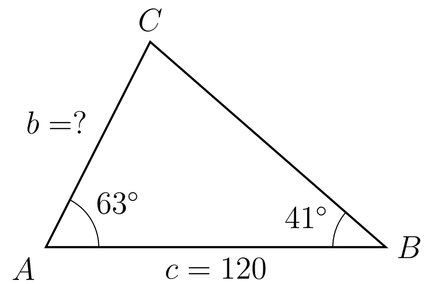
Answer: ________________

7. Surveying. A surveyor sights a rock across a ravine from two stations. The distance a shown is measured in metres, and the two marked angles are measured with a theodolite. Find the length b, in metres, to two decimal places.



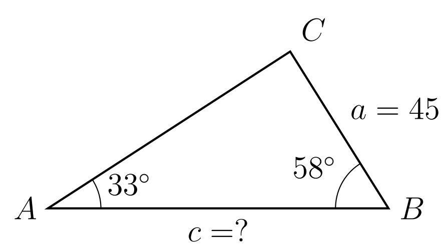
Answer: ________________ m

8. Baseline measurement. A straight baseline of length c metres is laid out between two stations A and B, and each station measures the angle to a transmitter at C. Find the distance b from station B to the transmitter, in metres.
Before computing, write down which angle is opposite the side you want — that is the pairing this problem is designed to test.



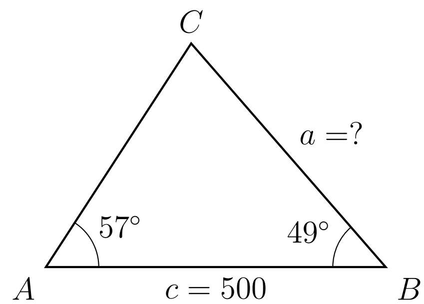
Answer: ________________ m

9. Two steps. Only side a and the angles A and B are given. Find side c. You will need the third angle before the law of sines can reach c; show that step explicitly.



Answer: ________________

10. Triangulation, then explain. Two observation posts A and B are c metres apart along a straight shoreline. Each post measures the angle to a buoy at C.
(a) Find the distance a from post B to the buoy, in metres.



Answer: ________________ m

(b) A classmate says "two angles and one side is not enough — you would need another measurement to be sure of the triangle." Explain why AAS or ASA data always determines exactly one triangle, and why two sides with a non-included angle (SSA) sometimes does not.